Year 9

REVISION PACK

Unit 3

Name: _____

Class: _____

Unit 3 · Arithmetic sequences (Lesson 3.1)

Worked example: nth term and testing a number

Here are the first four terms of an arithmetic sequence. 4 9 14 19

(a) Find an expression, in terms of n, for the nth term.

(b) Is 100 a term of the sequence? You must give a reason for your answer.

(a)	**Step 1**	Goes up 5 each time	$9 - 4 = 5$
	Step 2	Write the 5 times table	Term 4 9 14 19 $5n$ 5 10 15 20
	Step 3	Compare: each term is 1 less	$4 = 5 - 1$
	Step 4	Answer	$5n - 1$
(b)	**Step 1**	Make it equal to 100	$5n - 1 = 100$
	Step 2	Add 1	$5n = 101$
	Step 3	Divide by 5	$n = 20.2$
	Step 4	Answer	No, 20.2 is not a whole number

Practice: nth term and testing a number

1.1 Here are the first four terms of an arithmetic sequence: 5 9 13 17

(a) Find the nth term.

(b) Is 81 a term of the sequence? Give a reason.

1.2 Here are the first four terms of an arithmetic sequence: 2 8 14 20

(a) Find the nth term.

(b) Is 75 a term of the sequence? Give a reason.

1.3 Here are the first four terms of an arithmetic sequence: 20 17 14 11

(a) Find the nth term.

(b) Is 2 a term of the sequence? Give a reason.

1.4 Here are the first four terms of an arithmetic sequence: 3 10 17 24

(a) Find the nth term.

(b) Is 150 a term of the sequence? Give a reason.

1.5 Here are the first four terms of an arithmetic sequence: 11 16 21 26

(a) Find the nth term.

(b) Is 90 a term of the sequence? Give a reason.

1.6 Here are the first four terms of an arithmetic sequence: 40 34 28 22

(a) Find the nth term.

(b) Is 4 a term of the sequence? Give a reason.

Unit 3 · Quadratic sequences (Lesson 3.1)

Worked example: next term and nth term

Here are the first five terms of a quadratic sequence. 4 10 18 28 40

(a) Write down the next term of the sequence. (b) Find an expression, in terms of n, for the nth term.

(a)	Step 1	1st differences	Term 4 10 18 28 40 1st diff +6 +8 +10 +12
	Step 2	2nd difference	2nd diff +2 +2 +2
	Step 3	Next difference	$12 + 2 = 14$
	Step 4	Last term + next difference	$40 + 14 = 54$
(b)	Step 1	Half the 2nd difference	$2 \div 2 = 1$
	Step 2	So start with	n^2
	Step 3	Write n^2 under the terms	Term 4 10 18 28 40 n^2 1 4 9 16 25
	Step 4	Subtract: term − n^2	Subtract 3 6 9 12 15
	Step 5	This is the 3 times table	$3n$
	Step 6	Answer	$n^2 + 3n$

Practice: next term and nth term

2.1 Here are the first five terms of a quadratic sequence: 2 5 10 17 26

(a) Write down the next term. (b) Find the nth term.

2.2 Here are the first five terms of a quadratic sequence: 3 9 17 27 39

(a) Write down the next term. (b) Find the nth term.

2.3 Here are the first five terms of a quadratic sequence: 6 11 18 27 38

(a) Write down the next term. (b) Find the nth term.

2.4 Here are the first five terms of a quadratic sequence: 5 12 21 32 45

(a) Write down the next term. (b) Find the nth term.

2.5 Here are the first five terms of a quadratic sequence: 6 14 24 36 50

(a) Write down the next term. (b) Find the nth term.

2.6 Here are the first five terms of a quadratic sequence: 3 5 9 15 23

(a) Write down the next term. (b) Find the nth term.

Worked example: term-to-term rule and a later term

Here are the first four terms of a geometric sequence. 2 10 50 250

(a) Write down the term-to-term rule. (b) Work out the 6th term of the sequence.

(a)	Step 1	Term ÷ the term before	$10 \div 2 = 5$
	Step 2	Check	$50 \div 10 = 5$✓
	Step 3	Answer	Multiply by 5
(b)	Step 1	Keep multiplying by 5	Term: 2 (1st), 10 (2nd), 50 (3rd), 250 (4th), 1250 (5th), 6250 (6th)
	Step 2	5th term × 5	$1250 \times 5 = 6250$
	Step 3	Answer	6250

Practice: term-to-term rule and a later term

3.1 Here are the first four terms of a geometric sequence: 2 6 18 54

(a) Write down the term-to-term rule. (b) Work out the 6th term.

3.2 Here are the first four terms of a geometric sequence: 5 10 20 40

(a) Write down the term-to-term rule. (b) Work out the 8th term.

3.3 Here are the first four terms of a geometric sequence: 1 4 16 64

(a) Write down the term-to-term rule. (b) Work out the 5th term.

3.4 Here are the first four terms of a geometric sequence: 4 12 36 108

(a) Write down the term-to-term rule. (b) Work out the 5th term.

3.5 Here are the first four terms of a geometric sequence: 10 20 40 80

(a) Write down the term-to-term rule. (b) Work out the 6th term.

3.6 Here are the first four terms of a geometric sequence: 1 3 9 27

(a) Write down the term-to-term rule. (b) Work out the 7th term.

Unit 3 · Expanding brackets (Lesson 3.3)

Worked example: expand and simplify

Expand and simplify

(a) $4(x+3)-2(x-5)$ (b) $(x+5)(x-3)$ (c) $(2x+1)^2$

(a)	Step 1	Expand	$4x+12-2x+10$
	Step 2	Collect like terms	$4x-2x=2x, \quad 12+10=22$
	Step 3	Answer	$2x+22$
(b)	Step 1	Expand	$x^2-3x+5x-15$
	Step 2	Collect like terms	$-3x+5x=2x$
	Step 3	Answer	$x^2+2x-15$
(c)	Step 1	Expand	$(2x+1)(2x+1)=4x^2+2x+2x+1$
	Step 2	Collect like terms	$2x+2x=4x$
	Step 3	Answer	$4x^2+4x+1$

Practice: expand and simplify

4.1 Expand and simplify

(a) $3(x+4)-2(x+1)$ (b) $(x+2)(x+6)$ (c) $(x+5)^2$

4.2 Expand and simplify

(a) $5(x-2)-3(x-4)$ (b) $(x+5)(x-4)$ (c) $(x-3)^2$

4.3 Expand and simplify

(a) $2(3x-1)-4(x-3)$ (b) $(x-3)(x-7)$ (c) $(3x+2)^2$

4.4 Expand and simplify

(a) $6(x+1)-2(x+5)$ (b) $(x+3)(x+7)$ (c) $(x+4)^2$

4.5 Expand and simplify

(a) $3(2x-1)-2(x-3)$ (b) $(x-2)(x+6)$ (c) $(x-6)^2$

4.6 Expand and simplify

(a) $4(2x+3)-3(x-5)$ (b) $(x-5)(x-1)$ (c) $(2x-3)^2$

Unit 3 · Factorising (Lesson 3.4)

Worked example: factorise fully

Factorise fully

(a) $10x^2 - 15x$ (b) $x^2 + 7x + 10$ (c) $x^2 - 49$

(a)	**Step 1**	Find the HCF	$5x$
	Step 2	Divide each term by $5x$	$10x^2 \div 5x = 2x, \quad 15x \div 5x = 3$
	Step 3	Answer	$5x(2x - 3)$
(b)	**Step 1**	Multiply to 10, add to 7	$2 \times 5 = 10, \quad 2 + 5 = 7$
	Step 2	Answer	$(x + 2)(x + 5)$
(c)	**Step 1**	Write as squares	$x^2 - 7^2$
	Step 2	Answer	$(x + 7)(x - 7)$

Practice: factorise fully

5.1 Factorise fully

(a) $6x^2 + 9x$ (b) $x^2 + 7x + 12$ (c) $x^2 - 25$

5.2 Factorise fully

(a) $8x^2 - 12x$ (b) $x^2 + 8x + 12$ (c) $x^2 - 64$

5.3 Factorise fully

(a) $14x^2 - 21x$ (b) $x^2 + 3x - 10$ (c) $x^2 - 100$

5.4 Factorise fully

(a) $9x^2 + 12x$ (b) $x^2 + 7x + 6$ (c) $x^2 - 36$

5.5 Factorise fully

(a) $15x^2 - 10x$ (b) $x^2 - 2x - 15$ (c) $x^2 - 121$

5.6 Factorise fully

(a) $20x^2 + 8x$ (b) $x^2 - 7x + 12$ (c) $x^2 - 16$

Unit 3 · Solving quadratic equations (Lesson 3.5)

Worked example: solve the equations

Solve

(a) $x^2 - 2x - 15 = 0$ (b) $x^2 = 6x$

(a)	Step 1	Multiply to -15, add to -2	$3 \times -5 = -15, \quad 3 + (-5) = -2$
	Step 2	Factorise	$(x + 3)(x - 5) = 0$
	Step 3	Each bracket = 0	$x + 3 = 0$ or $x - 5 = 0$
	Step 4	Answer	$x = -3$ or $x = 5$
(b)	Step 1	Make one side 0	$x^2 - 6x = 0$
	Step 2	Factorise	$x(x - 6) = 0$
	Step 3	Answer	$x = 0$ or $x = 6$

Practice: solve the equations

6.1 Solve

(a) $x^2 - 5x + 6 = 0$ (b) $x^2 = 4x$

6.2 Solve

(a) $x^2 - 5x - 6 = 0$ (b) $x^2 = 7x$

6.3 Solve

(a) $x^2 + x - 12 = 0$ (b) $x^2 = 10x$

6.4 Solve

(a) $x^2 - 3x - 10 = 0$ (b) $x^2 = 3x$

6.5 Solve

(a) $x^2 + 7x + 6 = 0$ (b) $x^2 = 8x$

6.6 Solve

(a) $x^2 - 7x + 12 = 0$ (b) $x^2 = 5x$

Unit 3 · Problems with quadratics (Lesson 3.5)

Worked example: show that, then solve

A rectangle has length $(x + 2)$ cm and width $(x - 1)$ cm. The area of the rectangle is 18 cm².

(a) Show that $x^2 + x - 20 = 0$ (b) Work out the length of the rectangle.

(a)	**Step 1**	Length × width = area	$(x + 2)(x - 1) = 18$
	Step 2	Expand	$x^2 - x + 2x - 2 = 18$
	Step 3	Collect like terms	$x^2 + x - 2 = 18$
	Step 4	Subtract 18	$x^2 + x - 20 = 0$
(b)	**Step 1**	Factorise	$(x + 5)(x - 4) = 0$
	Step 2	Length is positive	$x = 4$
	Step 3	Length = x + 2	$\text{length} = 4 + 2 = 6$ cm

Practice: show that, then solve

7.1 A rectangle has length $(x + 4)$ cm and width $(x - 2)$ cm. Its area is 16 cm².

(a) Show that $x^2 + 2x - 24 = 0$ (b) Work out the length of the rectangle.

7.2 A rectangle has length $(x + 5)$ cm and width $(x - 1)$ cm. Its area is 27 cm².

(a) Show that $x^2 + 4x - 32 = 0$ (b) Work out the length of the rectangle.

7.3 A rectangle has length $(x + 6)$ cm and width $(x + 1)$ cm. Its area is 50 cm².

(a) Show that $x^2 + 7x - 44 = 0$ (b) Work out the width of the rectangle.

7.4 A rectangle has length $(x + 3)$ cm and width $(x - 2)$ cm. Its area is 24 cm².

(a) Show that $x^2 + x - 30 = 0$ (b) Work out the length of the rectangle.

7.5 A rectangle has length $(x + 7)$ cm and width $(x - 3)$ cm. Its area is 24 cm².

(a) Show that $x^2 + 4x - 45 = 0$ (b) Work out the width of the rectangle.

7.6 A rectangle has length $(x + 2)$ cm and width $(x + 1)$ cm. Its area is 56 cm².

(a) Show that $x^2 + 3x - 54 = 0$ (b) Work out the length of the rectangle.

Unit 3 · Perfect squares (Lesson 3.4)

Worked example: is it a perfect square?

(a) Show that $x^2 + 6x + 9$ is a perfect square. (b) Is $x^2 + 8x + 12$ a perfect square? Give a reason.

(a)	**Step 1**	Halve the number of x	$6 \div 2 = 3$
	Step 2	Square it	$3^2 = 9$, the same as the number term
	Step 3	Answer	$x^2 + 6x + 9 = (x + 3)^2$, so yes
(b)	**Step 1**	Halve the number of x	$8 \div 2 = 4$
	Step 2	Square it	$4^2 = 16$, but the number term is 12
	Step 3	Answer	No, $16 \neq 12$. It factorises to $(x + 2)(x + 6)$

Practice: is it a perfect square?

8.1 Is $x^2 + 10x + 25$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$

8.2 Is $x^2 - 8x + 16$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$

8.3 Is $x^2 + 6x + 8$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$

8.4 Is $x^2 - 14x + 49$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$

8.5 Is $x^2 + 12x + 30$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$

8.6 Is $x^2 - 2x + 1$ a perfect square? Give a reason for your answer.
If it is, write it in the form $(x + a)^2$ or $(x - a)^2$